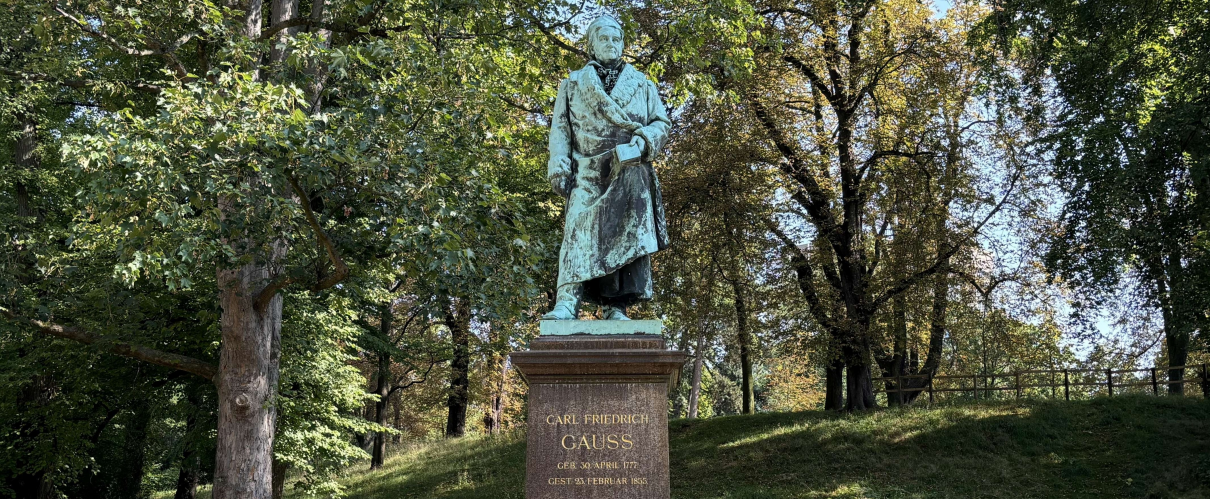




Engineering and Selecting Machine Learning Features for Feature Models

Master's Thesis | Student Research Competition, SE'26 | Raphael Dunkel (Kuitert, Sundermann, Tichy, Thüm) | February 26, 2026

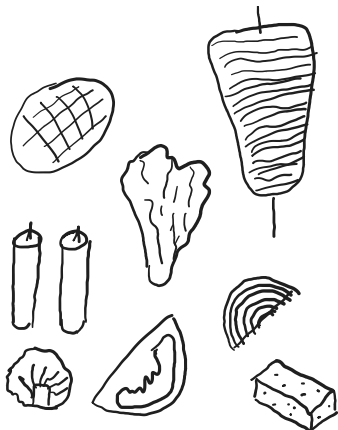


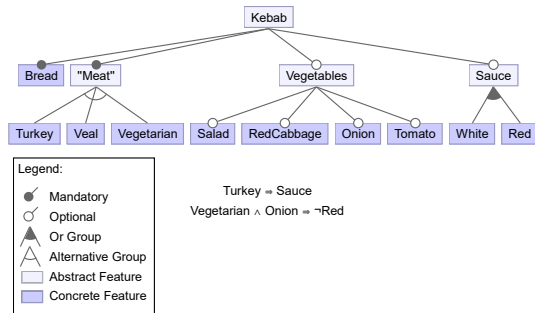
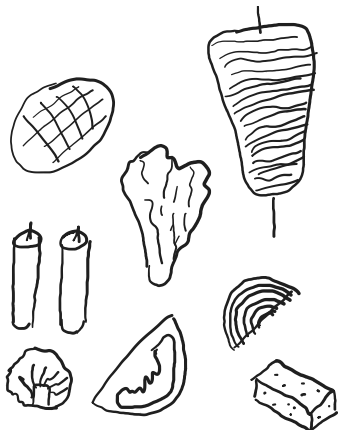
universität
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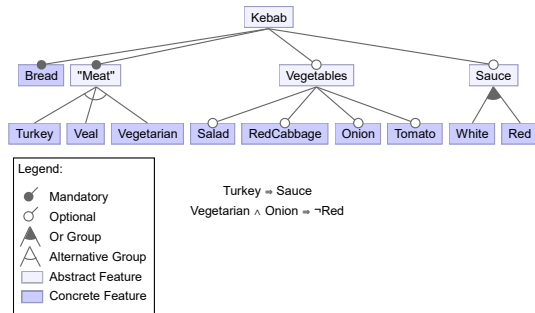
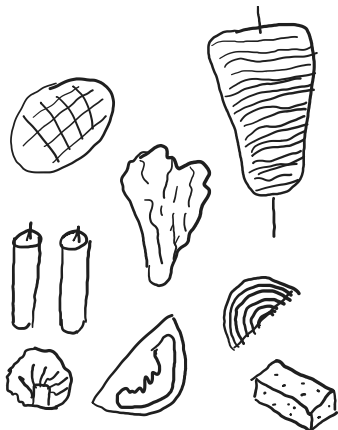


Software Engineering
Programming Languages

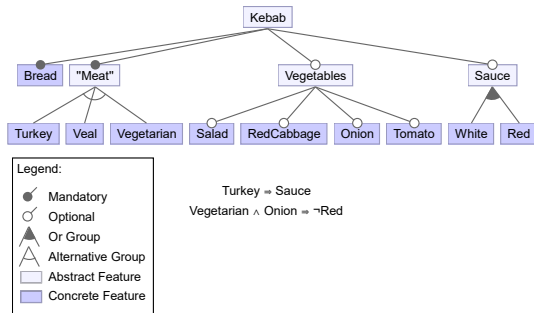
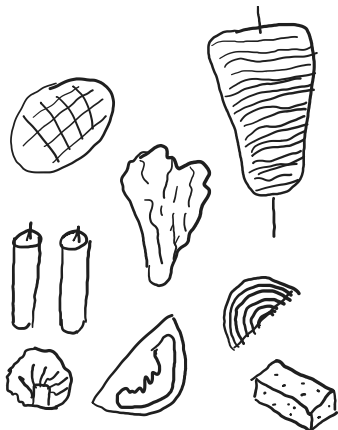




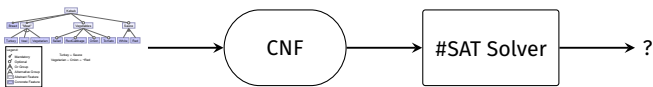


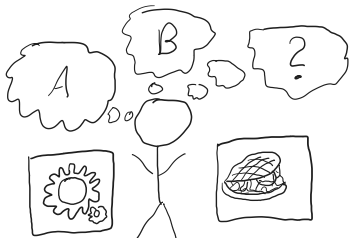


How many valid Kebabs are there?

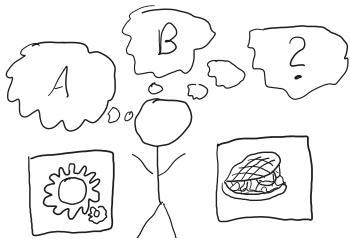


How many valid Kebabs are there?

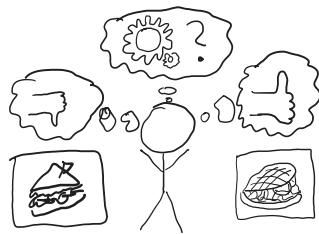




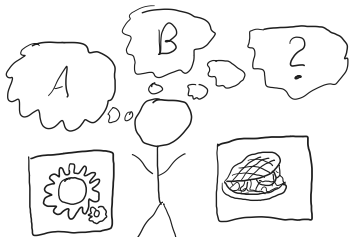
Which solver is best for my instance?



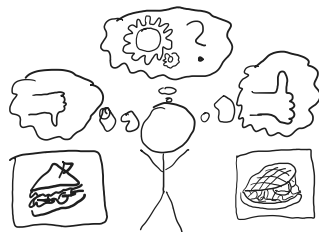
Which solver is best for my instance?



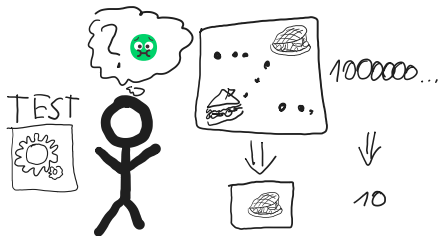
What makes feature models complex?



Which solver is best for my instance?



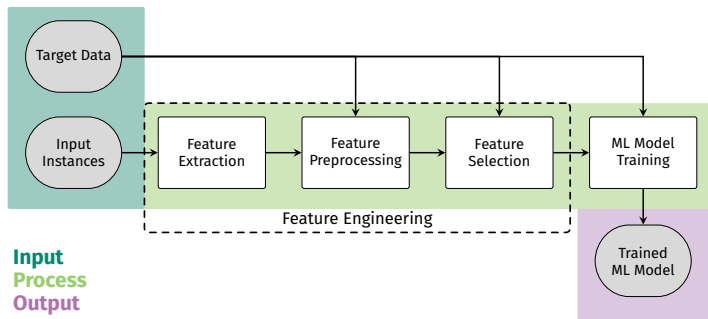
What makes feature models complex?



Which instances do I have to test?

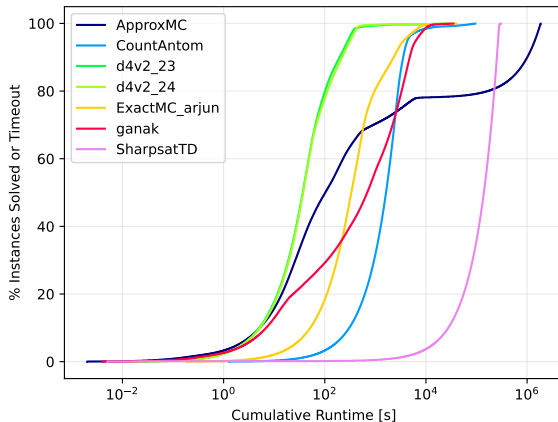
Contributions

1. ML Framework FE4FEMO



Contributions

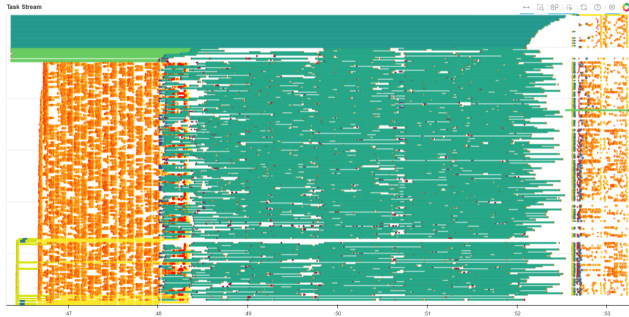
1. ML Framework FE4FEMO
2. New Data Sets for ML on Feature Models
 - Target Data
 - Extracted Features



Contributions

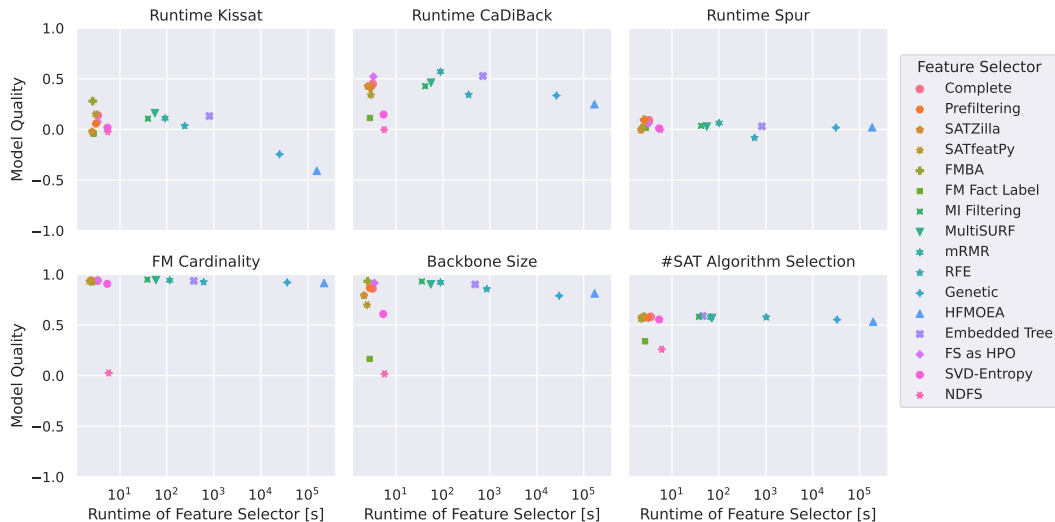
1. ML Framework FE4FEMO
2. New Data Sets for ML on Feature Models
 - Target Data
 - Extracted Features
3. Evaluation of ML Pipeline Configurations

$$6 \cdot 15 \cdot 6 \cdot 10 \cdot 150 \cdot 10 = 81000000s = 93.75d$$



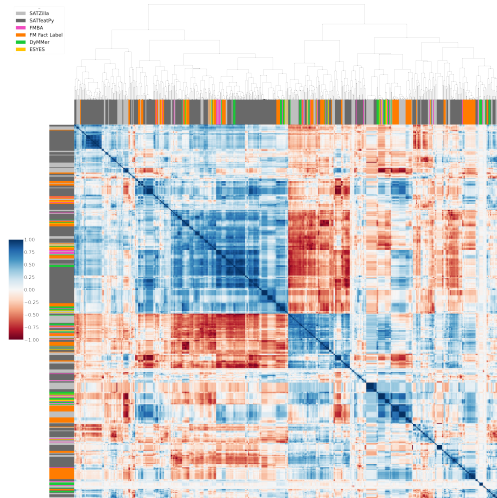
⇒ > 100 CPU years

Evaluation– Training Time Trade-Off

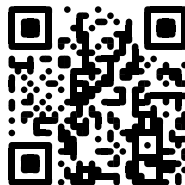
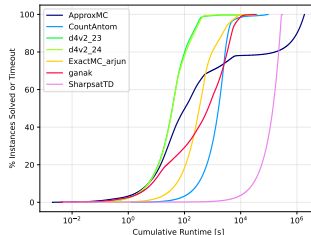
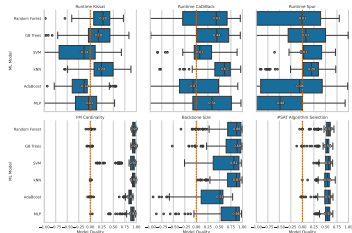
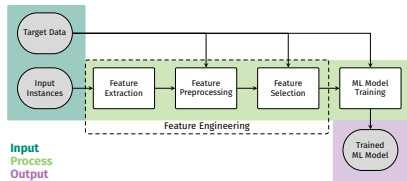


Possible Applications

- Per-instance machine learning for product line analysis applications
- In-depth feature space analysis
- Analysis of hyperparameter optimization impact
- Automated pipeline configuration
⇒ Auto-ML
- ...



What did we gain?



Appendix

Input Instances

Feature-Model Benchmark [2]

- biggest collection of feature models
- 2334 usable feature models
- 101 – 80,258 features
- various domains
- contains feature model histories
- UVL

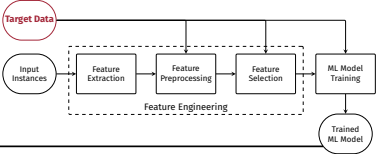
```
namespace TankWar

features
  TankWar {abstract true}
    mandatory
      Tank_Player1 {abstract true}
        alternative
          Black_P1
          Red_P1
          Red_Tier1_P1
          Yellow_Tier1_P1
          Blue_Tier2_P1
          Green_Tier2_P1
      Tank_Player2 {abstract true}
        alternative
          Black_P2
          Red_P2
          Red_Tier1_P2
          Yellow_Tier1_P2
          Blue_Tier2_P2
          Green_Tier2_P2
    mov {abstract true}
      alternative
        mov_0
        mov_1
        mov_2

[...]
```

[2] Sundermann et al., “Collecting Feature Models from the Literature: A Comprehensive Dataset for Benchmarking” (2024, ACM)

Target Data



Category	Task	Name	Extracted
SAT	Void	KISSAT	wallclock, memory
	Core-Dead	CADIBACK	wallclock, memory, backbone size
#SAT	Feature Model Cardinality	APPROXMC COUNTANTOM D4V2-23 D4V2-24 EXACTMC_ARJUN GANAK SHARPSATTD	wallclock, memory, algorithm selection, #SAT value
	Uniform Random Sampling	SPUR	wallclock, memory

FM Analysis [3]

⇒ Reusable Data Set for ML on Feature Models

[3] Sundermann et al., “On the Benefits of Knowledge Compilation for Feature-Model Analyses” (2024, Springer)

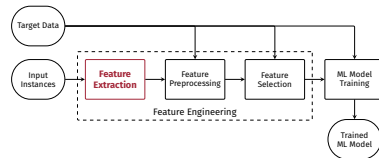
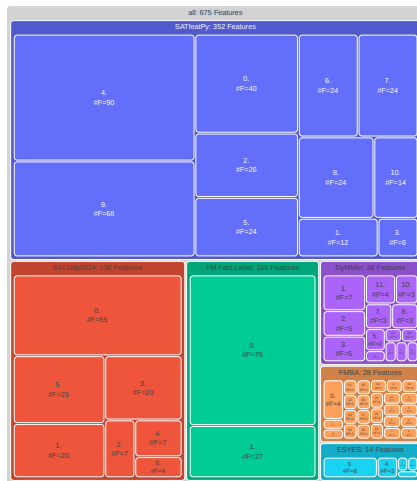
Feature Extraction

SAT:

- **SATZILLA** [4]
- **SATFEATPY** [5, 6]

FM Analysis:

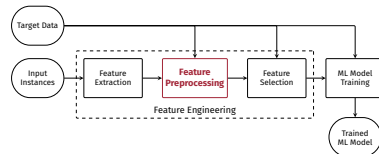
- **FMBA** [7, 8]
- **FM FACT LABEL** [9]
- **DYMMER** [10, 11]
- **ESYES** [12]



- Feature Groups
- Bug Fixes
- Optimizations: Caching, ...

⇒ Reusable Feature Values

Feature Preprocessing



MCAR assumption $\Rightarrow$ Mean Imputation

1. Feature Imputation
2. Feature Scaling
3. Variance and Correlation Filtering

$$Z = \frac{X - \mu}{\sigma}$$

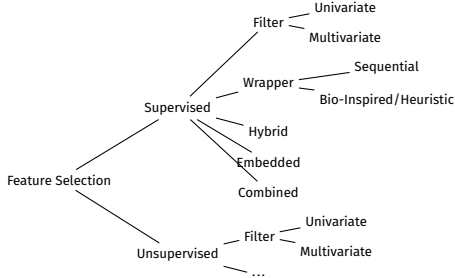
- Original Feature X
 - Mean μ
 - Standard Deviation σ
- Transformed Feature Z

Input : X :: Feature Matrix
Input : y :: Target Data
Output : X' :: Filtered Feature Matrix

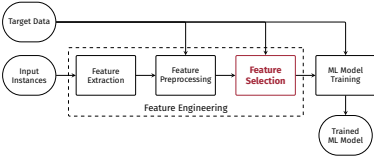
```

R ← ∅;
foreach  $x_i, x_j$  in  $\{(x_i, x_j) \mid \forall x_i, x_j \in X, i > j\}$  do
    if  $|r_s(x_i, x_j)| \geq 0.9$  then
        if  $|r_s(x_i, y)| \geq |r_s(x_j, y)|$  then
            if  $x_i \notin R$  then  $R \leftarrow R \cup \{x_i\}$ ;
        else
            if  $x_j \notin R$  then  $R \leftarrow R \cup \{x_j\}$ ;
return  $X \setminus R$ 
  
```

Feature Selection

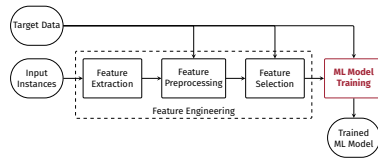
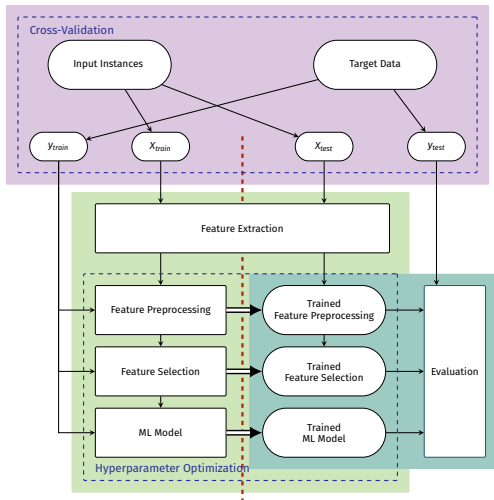


Adapted from [13, 14]



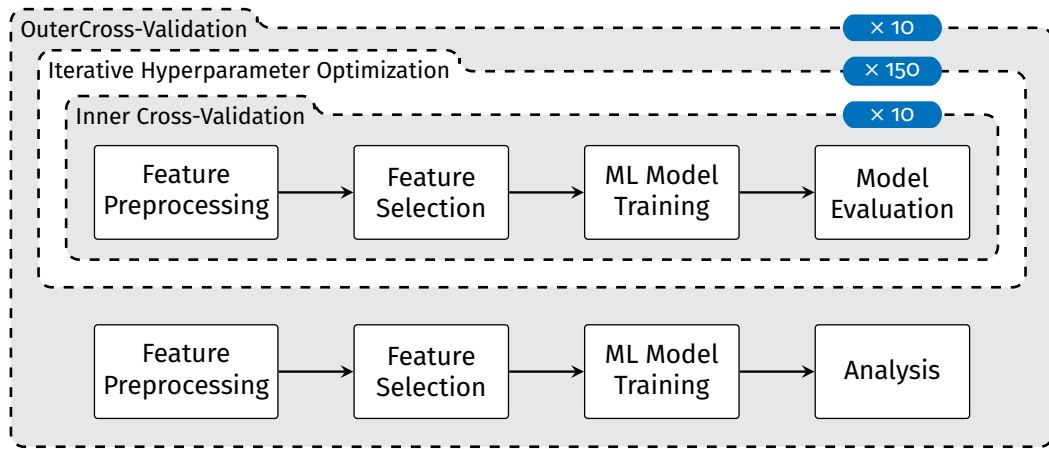
Category		Name	
Supervised	Filter	Univariate	Mutual Information Filtering
		Multivariate	MultiSURF
			mRMR
	Wrapper	Sequential	Recursive Feature Elimination
		Bio-Inspired / Heuristic	Genetic Algorithm
			Hybrid
	Embedded	Embedded Tree-based FS	
	Combined	Feature Selection as HPO	
Unsupervised	Filter	Univariate	SVD-Entropy Filtering
		Multivariate	NDFS

ML Model Training

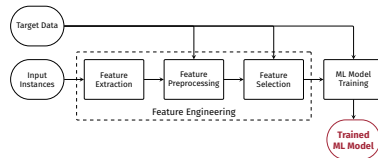


Model Name	Approaches
Random Forest	tree, ensemble, bootstrap aggregation
Gradient-Boosted Trees	tree, ensemble, boosting
Support Vector Machine	linear
<i>k</i> -Nearest Neighbors	clustering
AdaBoost	ensemble, boosting, (tree)
Multilayer Perceptron	neural network

Cross-Validation Strategy

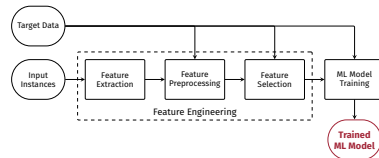


Evaluation – Research Questions



1. How well do different **ML pipeline configurations** especially **feature selection strategies**, perform for exemplary ML tasks on feature model analysis with regards to ...
 1. ...ML feature set **reduction**?
 2. ...feature selection **stability**?
 3. ...ML model **prediction quality**?
 4. What is the **trade-off** between all performance metrics?
2. How difficult are different **ML pipeline configurations**, especially **feature selection strategies**, to compute for exemplary ML tasks on feature model analysis with regards to ...
 1. ...feature selection algorithm **runtime**?
 2. ...active ML feature subset **computation time**?
 3. What is the **trade-off** between runtime, ML feature subset computation time, and ML model quality?
3. How is the combined **feature space** structured in terms of ...
 1. ...ML **feature correlation**?
 2. ...ML **feature importance**?

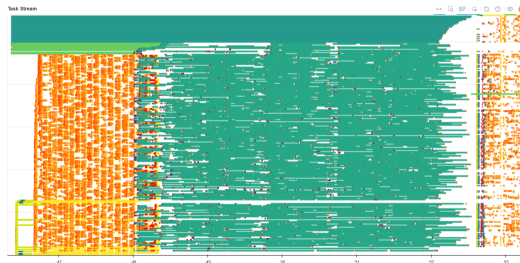
Evaluation – Experiment Execution



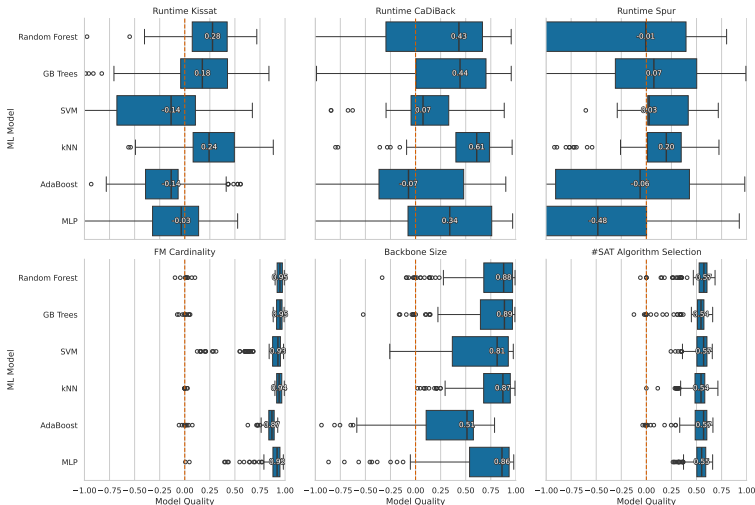
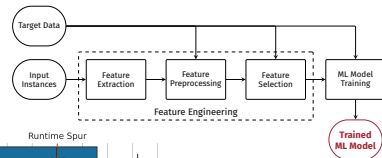
$$6 \cdot 15 \cdot 6 \cdot 10 \cdot 150 \cdot 10 = 81000000s = 93.75d$$

bwUniCluster 3.0+KIT-GFA-HPC 3

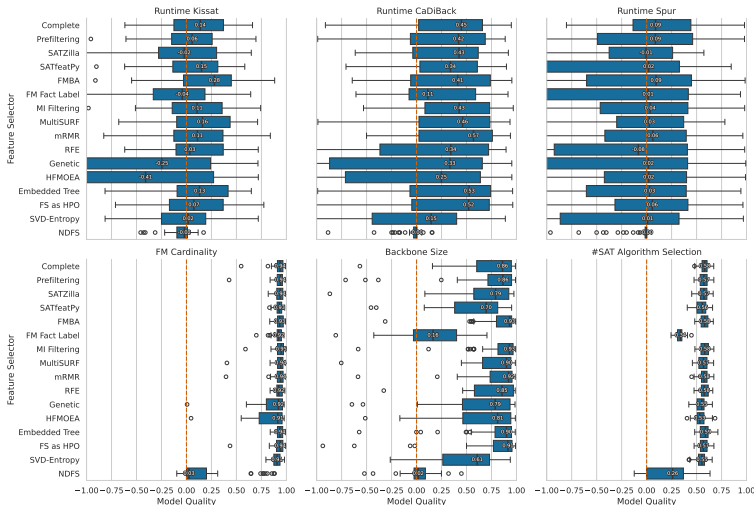
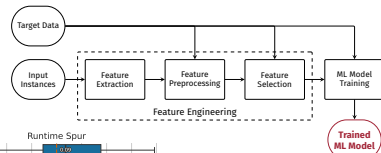
SLURM + DASK



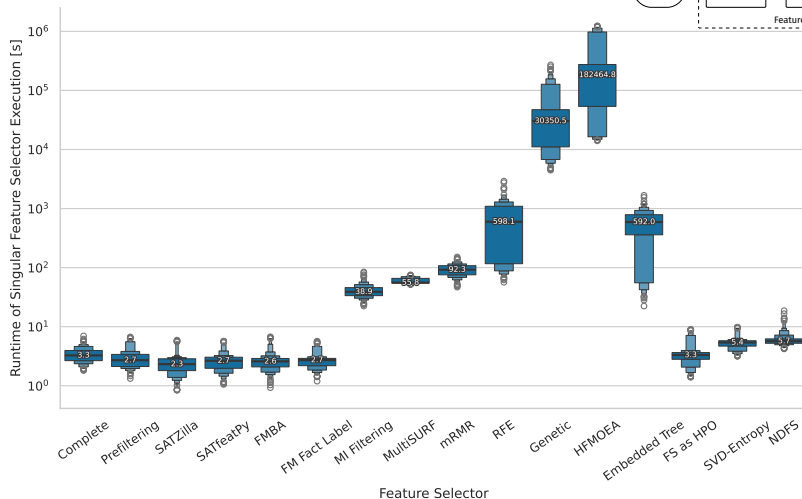
Evaluation – RQ 1.3: Performance



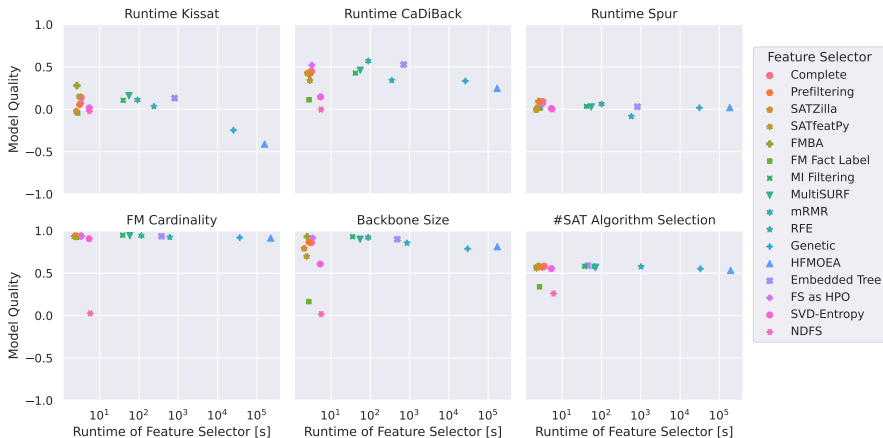
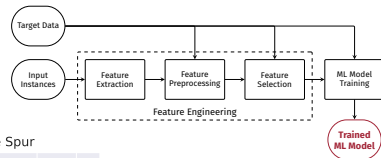
Evaluation – RQ 1.3: Performance



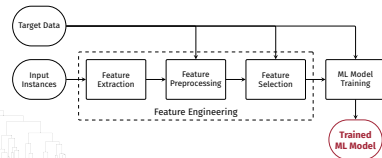
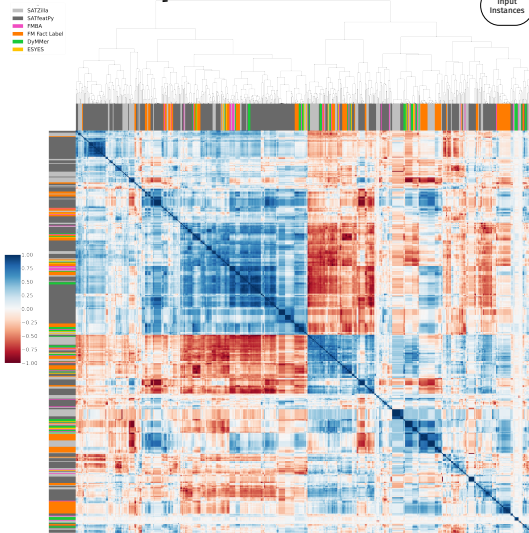
Evaluation – RQ 2.1: Computational Effort



Evaluation – RQ 2.3: Trade-Off



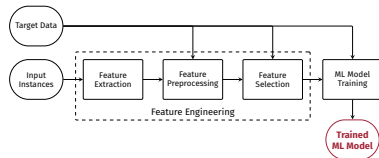
Evaluation – RQ 3.1: Feature Space



Evaluation – Threats to Validity

Internal Validity:

- Target Data Generation
 - Measurement errors
 - Right censoring
- Feature Extraction
 - Implementation errors
- Feature Preprocessing
 - Prefiltering can remove good features
 - Non-optimal imputation assumption
 - Downcasting
- Feature Selection
 - Implementation errors
- ML Model Training
 - Overfitting
 - Performance metrics



External Validity:

- Target Data Generation
 - Different Solvers could have different results
- Chosen Input Instances
 - Representative?
 - Groups in data set
 - Feature model size
- Selected ML Pipeline Configurations
 - Subset representative?
- No Established Evaluation Standards

Applications and Improvements

Improving Feature Extraction:

- Graph-based Features
- Solution Space Metrics
- Autoencoders
- Convolutional Neural Networks

Improving the Pipeline:

- Feature Transformations
- Imputation with MAR
- Feature Importance Analysis

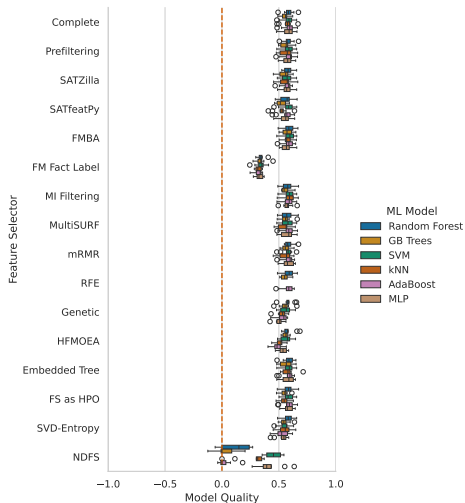
Analyzing FM Benchmarks:

- Clustering Analysis
- Benchmark Minimization
- Synthetic Benchmark Expansion

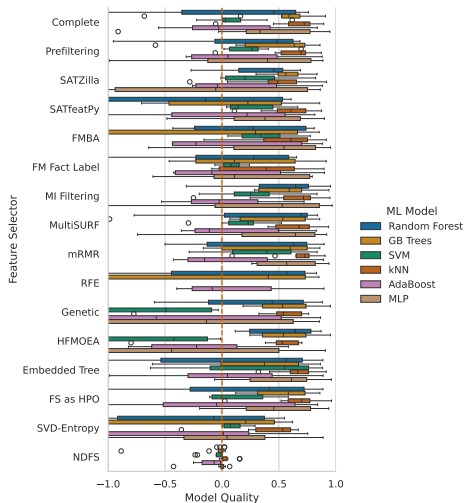
Solving Other Problems:

- #SAT Algorithm Selection
- FM Similarity
- FM Complexity Metrics

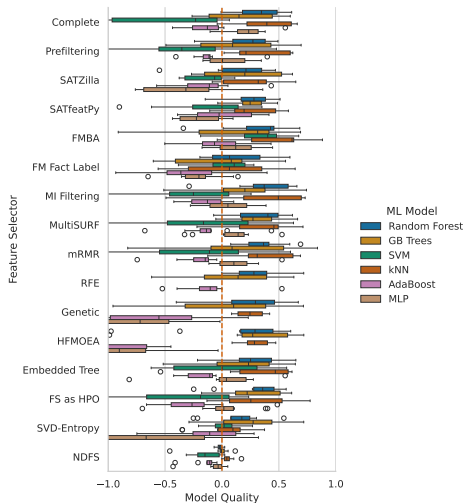
Model Performance – Algorithm Selection



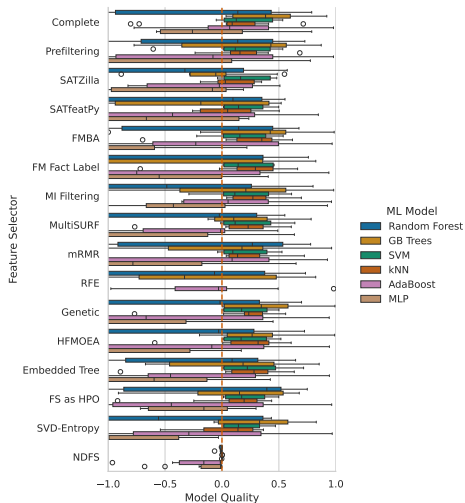
Model Performance – Runtime Backbone



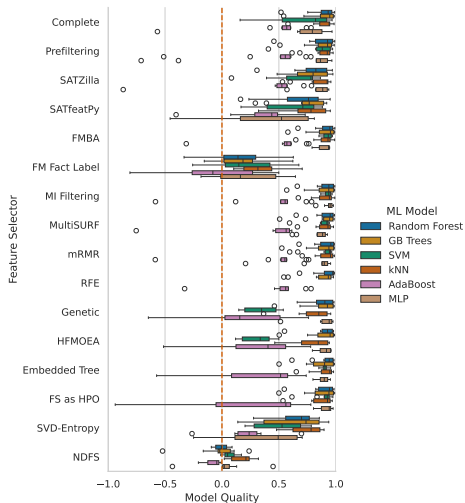
Model Performance – Runtime SAT



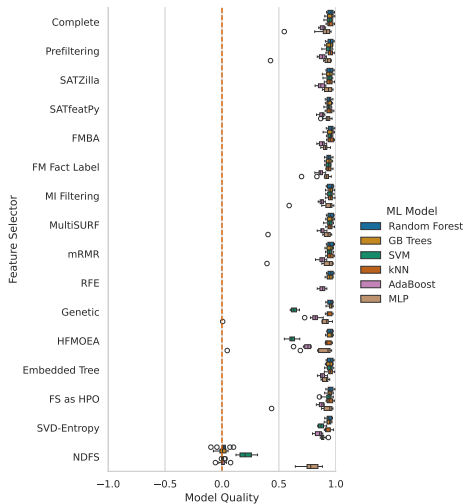
Model Performance – Runtime Spur



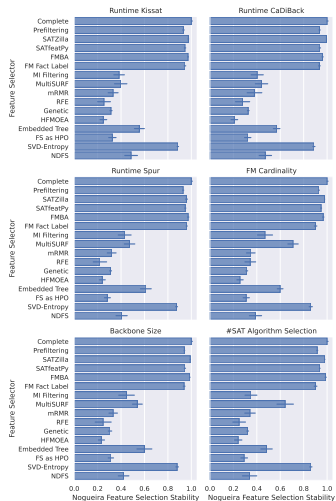
Model Performance – Value Backbone



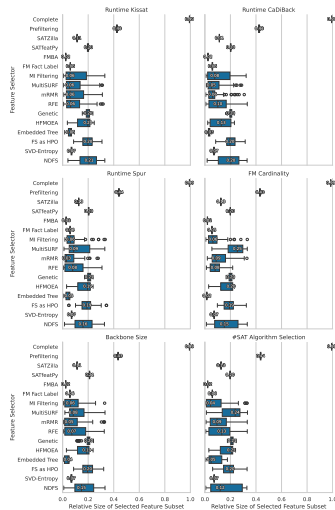
Model Performance – Value #SAT



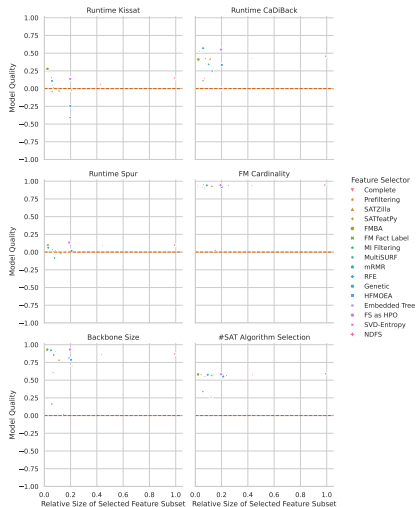
Stability



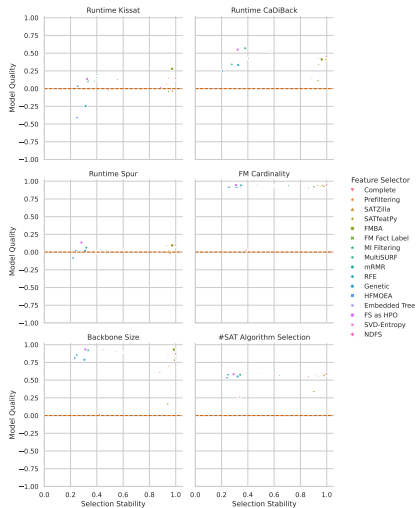
Reduction



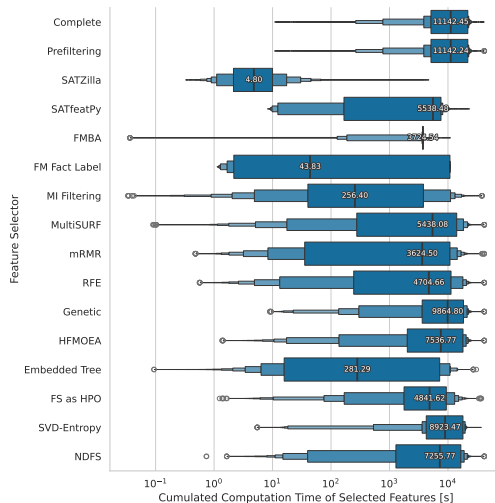
Trade-Off Quality Reduction



Trade-Off Quality Stability



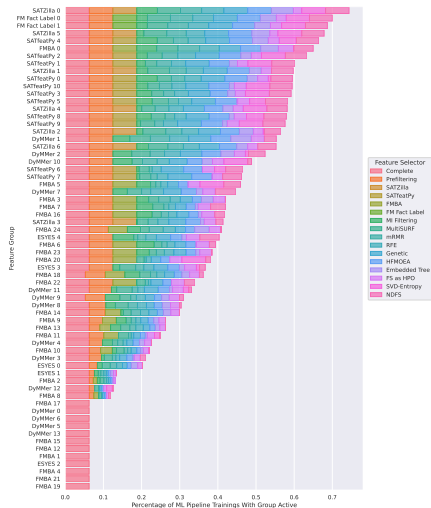
Feature Computation Time



Trade-Off Quality Both Times



Group Count



Feature Count

Feature Name	Feature Group	Usage Ratio
Number_CORE	FMBA 0	0.481362
AverageConstraintSize	FMBA 5	0.465054
lobjois-featuretime	SATZilla 6	0.463082
variable_alpha	SATfeatPy 3	0.461470
RatioOptionalFeatures	FMBA 0	0.447491
...		
Configuration_distribution/value	FM Fact Label 1	0

Bibliography I

- [1] V. Bolón-Canedo et al. “A Review of Microarray Datasets and Applied Feature Selection Methods”. 2014
- [2] Chico Sundermann et al. “Collecting Feature Models from the Literature: A Comprehensive Dataset for Benchmarking”. Sept. 2024
- [3] Chico Sundermann et al. “On the Benefits of Knowledge Compilation for Feature-Model Analyses”. Oct. 2024
- [4] Hadar Shavit and Holger H. Hoos. “Revisiting SATZilla Features in 2024”. 2024
- [5] Benjamin Provan-Bessell et al. *SATfeatPy – A Python-based Feature Extraction System for Satisfiability*. 2022
- [6] Marco Dalla et al. “SAT Feature Analysis for Machine Learning Classification Tasks”. 2023
- [7] Chico Sundermann et al. “Evaluating State-of-the-Art #SAT Solvers on Industrial Configuration Spaces”. Feb. 2024
- [8] Chico Sundermann et al. “Evaluating State-of-the-Art #SAT Solvers on Industrial Configuration Spaces”. Mar. 2023
- [9] Jos’e Miguel Horcas et al. “FM Fact Label”. 2025

Bibliography II

- [10] Anderson G. Uchôa et al. “DyMMer-NFP: Modeling Non-functional Properties and Multiple Context Adaptation Scenarios in Software Product Lines”. 2017
- [11] Carla I. M. Bezerra et al. “DyMMer: A Measurement-Based Tool To Support Quality Evaluation of DSPL Feature Models”. 2016
- [12] Sascha El-Sharkawy, Nozomi Yamagishi-Eichler, and Klaus Schmid. “Metrics for Analyzing Variability and its Implementation in Software Product Lines: A Systematic Literature Review”. 2019
- [13] Yvan Saeys, Iñaki Inza, and Pedro Larrañaga. “A Review of Feature Selection Techniques in Bioinformatics”. 2007
- [14] Saúl Solorio-Fernández, J. Ariel Carrasco-Ochoa, and José Fco. Martínez-Trinidad. “A Review of Unsupervised Feature Selection Methods”. 2020

Engineering and Selecting Machine Learning Features for Feature Models

1. Motivation

2. Contributions

3. Evaluation

4. Possible Applications

5. What did we gain?

6. Appendix